DAY 1 · AWS LINUX LAB

Systemd Service

Building a custom background service

Built `hello-loop.service` — a small custom systemd unit running a self-trimming logger script, from scratch.

Along the way, hit and fixed three real bugs (broken shebang, log permissions, restart rate-limiting) — exactly the kind of troubleshooting loop real systemd work involves.

Local — your machine · Remote — the server · Config file · Concept · Gotcha · Best practice · Paths

What we covered

- ✓ Wrote `hello-loop.sh` — a script logging a timestamp on an interval
- ✓ Placed it in `/usr/local/bin/` — the convention for sysadmin-authored scripts
- ✓ Built a systemd unit file with `[Unit]` / `[Service]` / `[Install]` sections
- ✓ Learned the `daemon-reload` → `enable` → `start` → `status` workflow
- ✓ Debugged a broken shebang line, a log permissions error, and a restart rate-limit lockout
- ✓ Refactored the log to self-trim to the last 5 lines, in memory, with no temp file

Full command sequence

Remote Create the script

```
sudo nano /usr/local/bin/hello-loop.sh   # create the script as root
```

/usr/local/bin/hello-loop.sh

```
#!/bin/bash
while true; do                                                          # loop forever
    echo "Hello from systemd at $(date)" >> /var/log/hello-loop.log     # >>: append a line
    echo "$(tail -n 5 /var/log/hello-loop.log)" > /var/log/hello-loop.log   # keep last 5 lines
    sleep 60                                                            # wait 60 seconds
done
```

```
sudo chmod +x /usr/local/bin/hello-loop.sh   # +x: make it executable
```

Remote Create the systemd unit file

```
sudo nano /etc/systemd/system/hello-loop.service   # create the unit file
```

/etc/systemd/system/hello-loop.service

```
[Unit]                                  # metadata + ordering
Description=Hello Loop Test Service     # shown in systemctl status
After=network.target                    # start after networking is up

[Service]                               # how to run it
ExecStart=/usr/local/bin/hello-loop.sh  # the command to run
Restart=always                          # restart it whenever it stops
User=adm-fkar                           # run as this user, not root

[Install]                               # how it hooks into boot
WantedBy=multi-user.target              # start in the normal boot state
```

Gotcha · Unit files only allow comments on their own line. Text after a value is read as part of the value, so the notes sit beside the file here instead of inside it.

Remote Enable, start, verify

```
sudo systemctl daemon-reload              # pick up the new unit file
sudo systemctl enable hello-loop.service  # start on boot
sudo systemctl start hello-loop.service   # start now
sudo systemctl status hello-loop.service  # check it's running
cat /var/log/hello-loop.log               # confirm it's writing
```

Bugs hit & fixed

Bug · broken shebang from an inline comment

CAUSE `#!/bin/bash # comment` — the kernel treats the whole shebang line as one interpreter path plus argument, so it tried to run bash with `# comment` and failed with exit code 127.

FIX Comments go on their own line — never on the shebang line itself.

Bug · log permission denied

CAUSE The service ran as `adm-fkar`, but `/var/log/` is root-owned — the user couldn't create files there.

FIX Create the log file as root, then hand that one file to the service user:

```
sudo touch /var/log/hello-loop.log                    # create the empty log file
sudo chown adm-fkar:adm-fkar /var/log/hello-loop.log  # give it to the service user
```

Bug · restart rate-limit lockout

CAUSE After several failed starts, systemd refused further attempts (“Start request repeated too quickly”).

FIX Clear the failed state, then start again:

```
sudo systemctl reset-failed hello-loop.service  # clear the crash counter
sudo systemctl start hello-loop.service         # try again
```

Where things live

What	Path
Custom scripts (sysadmin-authored)	`/usr/local/bin/`
Distro-managed binaries	`/usr/bin/`
Systemd unit files	`/etc/systemd/system/`
hello-loop's log	`/var/log/hello-loop.log`

Best practices

No comments on the shebang
The kernel reads the whole line as the interpreter path.

Own the log file up front
Give a service's log file explicit ownership before the service ever runs, not after it fails.

Unstick a crash loop
Run `systemctl reset-failed` when a crash-looped service refuses to restart.

Trim in memory
Prefer `echo "$(tail ...)" > file` over a temp-file-and-move dance when the directory isn't writable by the service user.

Always daemon-reload
Run it after creating or editing any unit file.

Scripts in /usr/local/bin
Keep your own scripts apart from distro binaries in `/usr/bin/`.

Part of the AWS/Linux sysadmin lab · see day1-naming-conventions for the full naming reference